

IET, DAVV, Indore
B.E. II (ETC)
Class Test I (Aug 2023)
ATR3C1: Applied Mathematics-III

Time: 70 min

Max Marks: 20

Note: Attempt any FOUR questions. All questions carry equal marks.

- Q.1. Find positive root of $x^3 - x = 1$, lie between (1, 2). Perform four iterations of method (05)
(Bisection or Newton Raphson method).
- Q.2. Use Euler's modified formula to solve $\frac{dy}{dx} = 1 - 2xy$, given $y = 0$ at $x = 0$. Evaluate $x = 0$ to (05)
 $x = 0.4$, taking interval $h = 0.2$ correct upto 2 decimal places.
- Q.3. Solve the given system of linear equation using Jacobi method correct upto two decimal (05)
places.
 $x + y + 10z = 35.61$
 $10x + y - z = 11.19$
 $x + 10y + z = 28.08$
- Q.4. Find $y(0.2)$ for $\frac{dy}{dx} = -y$, $x_0 = 0$, $y_0 = 1$, using Taylor series method. (05)
- Q.5. Find the solution of an equation $2x^3 - 4x + 1$, using Newton's Backward Difference formula. (05)
Find $f(x)$ at $x = 3.75$.

x	2	2.5	3	3.5	4
F(x)	9	22.25	43	72.75	113
